



# LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



## GNSS Interference Aviation Operational Cyber Risk Daily Review

### Threat Report

|                       |                 |
|-----------------------|-----------------|
| <b>Classification</b> | TLP:CLEAR       |
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TLP:CLEAR | Lost Boys Cyber | GNSS Interference Aviation Operational Cyber Risk Daily Review - Threat Report

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# GNSS Interference Aviation Operational Cyber Risk Daily Review

## 1. Executive Summary

EASA, FAA, IATA, IFALPA, and related aviation sources describe GNSS radio-frequency interference, jamming, and spoofing as a persistent operational risk. While not always a classic enterprise intrusion, GNSS interference has cybersecurity consequences for monitoring, operational continuity, sensor trust, and crisis communications.

This report was selected for the 2026-07-31 UTC daily Lost Boys Cyber production run because the item is recent or materially relevant to defenders within the last 24-72 hours; where fewer fresh primary sources were available, the window was expanded to seven days and documented in the source review.

## 2. Intelligence Requirements

| Question                               | Current Answer                                                                                                                                                                                                                                                                                                         | Confidence |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| What happened?                         | Aviation operators should treat GNSS interference as a cross-domain cyber-physical risk. Security teams can support operations by monitoring disruptions, protecting reporting channels, validating satellite-dependent integrations, and rehearsing degraded-mode procedures with flight operations and safety teams. | High       |
| Who should prioritize review?          | Flight operations, navigation/surveillance, satellite-dependent aviation services                                                                                                                                                                                                                                      | Medium     |
| What is the immediate defender action? | Validate exposure, tune detections, run hunts, and confirm response ownership.                                                                                                                                                                                                                                         | High       |

## 3. Source Review & Confidence

| Source                                        | Contribution                                                      | Confidence |
|-----------------------------------------------|-------------------------------------------------------------------|------------|
| EASA GNSS interference SIB update             | Primary aviation safety update on GNSS outage/alteration risks.   | High       |
| FAA GPS/GNSS Interference Resource Guide      | Operational guide for jamming/spoofing trends and procedures.     | High       |
| IATA GNSS RFI safety risk assessment          | Industry risk assessment for GNSS RFI impacts.                    | High       |
| IFALPA GNSS RFI and signal manipulation paper | Pilot/operational perspective on escalating jamming and spoofing. | High       |

## 4. Threat Activity Overview

Aviation operators should treat GNSS interference as a cross-domain cyber-physical risk. Security teams can support operations by monitoring disruptions, protecting reporting channels, validating satellite-dependent integrations, and rehearsing degraded-mode procedures with flight operations and safety teams.

## 5. Affected Sectors and Exposure

| Sector / Asset Group | Exposure Consideration | Analyst Note |
|----------------------|------------------------|--------------|
|----------------------|------------------------|--------------|

|                                                                                   |                                                                           |                                                                                                   |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Flight operations, navigation/surveillance, satellite-dependent aviation services | Current reporting maps directly or indirectly to this environment class.  | Prioritize assets with external exposure, privileged identity reach, or operational dependencies. |
| Third-party providers                                                             | Shared platforms and managed services can carry the operational impact.   | Request patch, detection, and incident-notification evidence.                                     |
| Remote workforce / identity plane                                                 | Many campaigns begin with credential, phishing, or trusted-channel abuse. | Review MFA, conditional access, impossible travel, and helpdesk reset controls.                   |

## 6. Aviation and Aerospace Sector Impact

| Segment                                                                           | Operational Relevance                                                                                                                                              | Likely Exposure Path                                                                                                                           | Defender Action                                                                                                                            |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Flight operations, navigation/surveillance, satellite-dependent aviation services | Current source material has aviation/aerospace operational relevance and may affect safety-adjacent business continuity, regulated operations, or mission support. | Exposure may occur through internet-facing services, trusted suppliers, identity workflows, operational IT, or sector-specific communications. | Validate ownership, patch and monitor relevant systems, rehearse degraded operations, and confirm third-party incident notification paths. |
| Third-party and supply chain                                                      | Aviation outcomes often depend on shared service providers, airport/airline vendors, OEM/MRO integrations, and cloud/SaaS platforms.                               | Vendor compromise, unavailable managed platforms, delayed disclosure, or shared credentials can cascade beyond one organization.               | Require vendor attestations, segment integrations, maintain offline continuity plans, and define escalation contacts before an incident.   |

## 7. Tactics, Techniques, and Procedures

| Tactic / Phase                | Observed Technique                                                            | Evidence                                                                                                                                       |
|-------------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Initial Access                | Phishing, exploit public-facing application, or supply-chain trust abuse      | Source reporting describes campaign or sector activity that begins through exposed services, trusted communications, or software dependencies. |
| Execution                     | Script, malware, or administrative-tool execution                             | Defender review should focus on process creation, browser-mediated execution, package scripts, or suspicious service-account activity.         |
| Persistence / Defense Evasion | Valid accounts, relay infrastructure, tool renaming, or trusted process abuse | Activity may blend into normal administrative or browser/cloud traffic.                                                                        |
| Impact                        | Data theft, operational disruption, ransomware, or mission degradation        | Sector impact depends on operational criticality and third-party dependencies.                                                                 |

## 8. Infrastructure and Indicators

| Indicator / Infrastructure                             | Type       | Operational Use                                                                                                       |
|--------------------------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------|
| Source-reported domains, IPs, hashes, or package names | Observable | No hard IOC was safely extracted during automated run unless listed in source material; analysts should enrich source |

|                                                                                         |                   |                                                                        |
|-----------------------------------------------------------------------------------------|-------------------|------------------------------------------------------------------------|
|                                                                                         |                   | pages before blocking.                                                 |
| Abnormal sign-in, script execution, web-service child processes, package publish events | Behavioral signal | Preferred for durable detection when hard IOCs are absent or volatile. |

## 9. Detection Engineering

### 9.1. Detection Summary

| Signal                                                             | Telemetry Source                         | Analytic Goal                                                                 |
|--------------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------------------------|
| Suspicious command execution after trusted-app or browser activity | EDR process telemetry                    | Identify post-compromise execution hidden behind legitimate parent processes. |
| Unusual outbound traffic from servers or developer endpoints       | DNS, proxy, firewall, EDR network events | Surface payload retrieval, C2, or data staging.                               |
| Identity anomalies following phishing or supply-chain exposure     | Entra ID, Okta, VPN, SSO logs            | Find compromised accounts before lateral movement.                            |

### 9.2. Sigma / Detection Content

```

title: Daily Threat Review Suspicious Trusted Process Follow-On Activity
status: experimental
description: Detects suspicious command or scripting tools launched from browsers, collaboration
clients, package managers, or service parents during daily threat-review windows.
logsource:
  product: windows
  category: process_creation
detection:
  trusted_parent:
    ParentImage|contains:
      - chrome.exe
      - msedge.exe
      - teams.exe
      - npm.cmd
      - node.exe
      - w3wp.exe
      - outlook.exe
  suspicious_child:
    Image|endswith:
      - \powershell.exe
      - \pwsh.exe
      - \cmd.exe
      - \wscript.exe
      - \cscript.exe
      - \rundll32.exe
      - \regsvr32.exe
  condition: trusted_parent and suspicious_child
fields:
  - UtcTime
  - Computer
  - ParentImage
  - Image
  - CommandLine
level: medium

```

### 9.3. Hunt Query

```

DeviceProcessEvents
| where Timestamp > ago(14d)
| where InitiatingProcessFileName in~ ("chrome.exe", "msedge.exe", "teams.exe", "outlook.exe",

```

```
"npm.cmd", "node.exe", "w3wp.exe")
| where FileName in~ ("powershell.exe", "pwsh.exe", "cmd.exe", "wscript.exe", "cscript.exe",
"rundll32.exe", "regsvr32.exe")
| summarize count(), min(Timestamp), max(Timestamp) by DeviceName, InitiatingProcessFileName, FileName,
AccountName, ProcessCommandLine
| order by count_ desc
```

## 10. Threat Hunting

### 10.1. Hunt Hypothesis

If the reported activity is present locally, telemetry should show anomalous identity behavior, suspicious trusted-process child execution, abnormal network destinations, or unusual package/script activity around the initial access window.

### 10.2. Hunt Query

```
SigninLogs
| where TimeGenerated > ago(14d)
| where ResultType != 0 or RiskLevelAggregated in ("medium", "high") or ConditionalAccessStatus ==
"failure"
| summarize Events=count(), FirstSeen=min(TimeGenerated), LastSeen=max(TimeGenerated) by
UserPrincipalName, IPAddress, AppDisplayName, RiskLevelAggregated, ConditionalAccessStatus
| order by Events desc

DeviceNetworkEvents
| where Timestamp > ago(14d)
| where InitiatingProcessFileName in~ ("chrome.exe", "msedge.exe", "teams.exe", "node.exe", "npm.exe",
"python.exe", "powershell.exe")
| where RemoteUrl !endswith ".microsoft.com" and RemoteUrl !endswith ".windows.com"
| summarize Connections=count(), FirstSeen=min(Timestamp), LastSeen=max(Timestamp) by DeviceName,
InitiatingProcessFileName, RemoteUrl, RemoteIP
| order by Connections desc
```

## 11. Risk Assessment

| Dimension  | Assessment                                                                                                                |
|------------|---------------------------------------------------------------------------------------------------------------------------|
| Likelihood | High priority based on current reporting, sector exposure, and adversary utility.                                         |
| Impact     | Potential business disruption, credential theft, data exposure, or operational degradation depending on local dependency. |
| Confidence | High for prioritization; local exposure requires validation.                                                              |

## 12. Recommended Actions

| Priority | Owner                        | Action                                                                                               |
|----------|------------------------------|------------------------------------------------------------------------------------------------------|
| Critical | SOC                          | Run hunts across the prior 14 days and open cases for confirmed local hits.                          |
| High     | Engineering                  | Validate patch, identity, package, and third-party controls relevant to the affected exposure path.  |
| High     | Incident response            | Prepare containment steps and preserve logs for high-value systems.                                  |
| Medium   | Governance/vendor management | Request third-party evidence for aviation, supply-chain, identity, and managed-service dependencies. |

## 13. References

- [1] EASA GNSS interference SIB update: <https://www.easa.europa.eu/en/newsroom-and-events/news/easa-updates-safety-information-bulletin-gnss-interference>
- [2] FAA GPS/GNSS Interference Resource Guide:  
[https://www.faa.gov/about/office\\_org/headquarters\\_offices/avs/offices/afx/afs/afs400/afs410/GNSS/GPS\\_GNSS\\_Interference\\_Resource\\_Guide.pdf](https://www.faa.gov/about/office_org/headquarters_offices/avs/offices/afx/afs/afs400/afs410/GNSS/GPS_GNSS_Interference_Resource_Guide.pdf)
- [3] IATA GNSS RFI safety risk assessment: [https://ic.iata.org/sites/default/files/iata\\_sih\\_document\\_attachment/IATA+Safety+Risk+Assessment+-+GNSS+Interference+V6.pdf](https://ic.iata.org/sites/default/files/iata_sih_document_attachment/IATA+Safety+Risk+Assessment+-+GNSS+Interference+V6.pdf)
- [4] IFALPA GNSS RFI and signal manipulation paper: <https://www.ifalpa.org/wp-content/uploads/2026/05/26sab01-gnss-radio-frequency-interference-and-signal-manipulation.pdf>